



University of Pisa

Master's Degree in Computer Science

Curriculum: Artificial Intelligence



Machine Learning for Satellite Link Fade Nowcasting

Giuseppe Gabriele Russo

Experimental thesis carried out at M.B.I. S.r.l.

Supervisor
Davide Bacciu

Co-supervisor
Giovanni Scognamiglio

Industrial Context and Operational Problem

Experimental work at M.B.I.

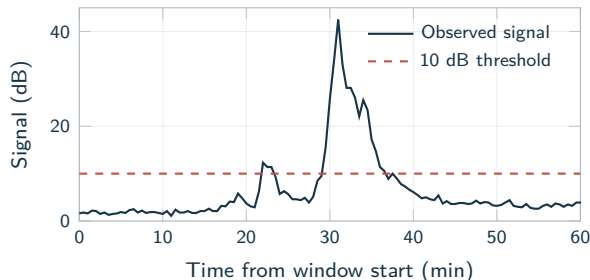
Real satellite fade and beacon recordings from the Fucino ground station.

The operational question

Will the fade persist long enough to justify a backup-link switch?

Switching late leaves the link exposed. Reacting to every fluctuation causes unnecessary activation.

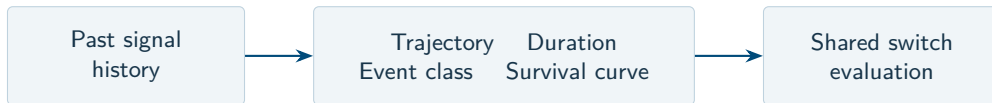
Anticipate persistence, rather than only detect a threshold crossing.



Excerpt from the held-out source; future samples are unavailable to the predictor at decision time.

Research Objective and Contribution

Which predictive formulation supports useful switch decisions from signal history alone?



- Compare four task formulations using established model families.
- Separate predictive accuracy from operational switch behavior.
- Make decision coverage explicit when comparing different tasks.

The contribution is a decision-level comparison, not a new neural architecture.

Data and Experimental Setting

Shared data preparation

- Signal-only inputs; 30-second sampling.
- 30-observation context: approximately 15 minutes.
- Timestamp validation and conservative gap handling.
- Task-specific targets and eligibility rules.

Small-gap policy: at most two inserted samples within a 90-second gap; no interpolation across long acquisition gaps.

Development sources

Chronological training / validation splits.

Validation-based model selection.

External test source

`fc-uplink-fade.csv`

Excluded from fitting and hyperparameter search.

Chronos: zero-shot. Survival switch threshold: exploratory.

A common signal source and split protocol, with different prediction targets.

Four Predictive Formulations

Formulation	Question at time t	Output
Autoregressive	What are the next ten signal values?	Future trajectory
Current-level	How long until the signal falls below $S_t - 0.5$ dB?	Duration estimate
Long-fade	Does the entire grouped event last at least 300 seconds?	Class probability
Survival	Will the current fade remain active for more than u seconds?	Survival curve $S(u X_t)$

All use past signal information; their targets are not interchangeable.

Perfect Switch and Evaluation

Perfect Switch

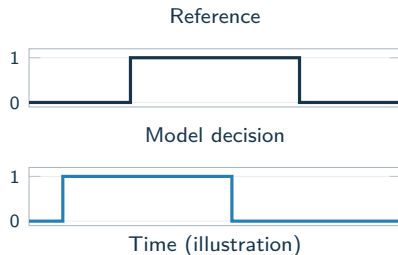
An offline reference using the true signal and future persistence.

An evaluation oracle, not a deployable predictor.

Complementary measurements

- Precision, recall, F1, balanced accuracy.
- Active duration and switch intervals.
- Timing and missed activity.

Measure what the decision does, not only how accurate the prediction is.



Shared Switch Post-Processing

1. Raw decision

Apply the task-specific conversion rule.

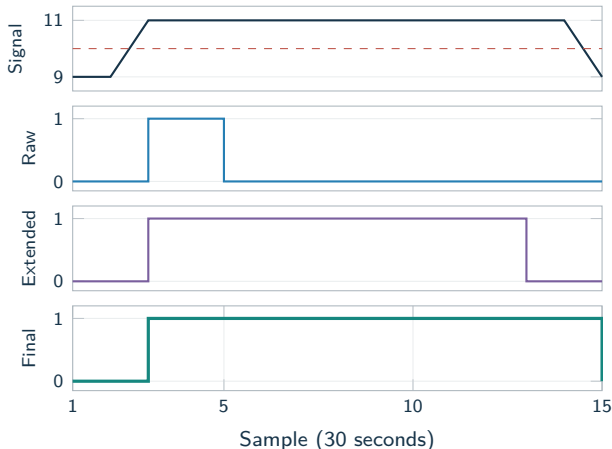
2. Minimum active interval

Extend short positive islands to the configured ten-sample length.

3. Hold while above threshold

Keep an already active switch on while the current signal remains ≥ 10 dB.

This modifies the binary decision, not the input signal.



Autoregressive Forecasting

Predict the next ten values

$$X_t = (S_{t-29}, \dots, S_t) \longrightarrow (\hat{S}_{t+1}, \dots, \hat{S}_{t+10})$$

GRU sequence-to-vector and encoder-decoder;
PatchTST; XGBoost per horizon; Chronos
zero-shot.

Decision before shared post-processing

$$a_t = 1 \quad \text{if } \hat{S}_{t+k} \geq 10 \quad \forall k = 1, \dots, 10$$

Raw and context-standardized trainable variants. Chronos uses raw contexts; L120 is an additional experiment.

External test, native L30 grid

Model	RMSE ↓	F1 ↑
	(dB)	(switch)
GRU S2V raw	2.852	0.674
PatchTST raw	3.088	0.808

PatchTST provides the stronger switch decision despite a higher trajectory RMSE.

The best forecasting score does not select the best operational policy.

Current-Level Persistence

Predict time below the current level

$$R_t^{\text{level}} = \inf\{u > 0 : S_{t+u} < S_t - 0.5\}$$

Learnable shapelets with MLP, convolutional or Transformer heads; XGBoost baseline. Relative histories plus scalar signal context.

Decision before shared post-processing

$$a_t = 1 \quad \text{if } S_t \geq 10 \text{ and } \hat{R}_t^{\text{level}} \geq 300 \text{ s}$$

XGBoost: external test, native grid

Duration MAE ↓	586.9 s
----------------	---------

Duration median AE ↓	27.2 s
----------------------	--------

Switch precision ↑	1.000
--------------------	-------

Switch recall ↑	0.822
-----------------	-------

Switch F1 ↑	0.902
-------------	-------

Shapelet variants never activate the final switch: their duration estimates do not satisfy the operational gate.

A conservative XGBoost policy succeeds where the shapelet duration models fail.

Long-Fade Detection

Classify the entire grouped event

$$p_t = P(D_{\text{event}} \geq 300 \text{ s} \mid X_t)$$

XGBoost, causal TCN and shapelet convolution classifier.

Decision before shared post-processing

$$a_t = 1 \quad \text{if } p_t \geq 0.5$$

The label remains positive throughout a long grouped event, including near its end. It does not describe the remaining duration.

TCN: external test, native grid

Classification AUPRC $\uparrow$	≈ 1.000
---------------------------------	-----------------

Switch precision $\uparrow$	0.540
-----------------------------	-------

Switch recall $\uparrow$	1.000
--------------------------	-------

Switch F1 $\uparrow$	0.702
----------------------	-------

Active duration	445 min
-----------------	---------

Classification is evaluated on an event-conditioned, strongly imbalanced domain.

Near-perfect event classification does not imply accurate switch timing.

Survival Persistence

Predict residual-event uncertainty

$$S(u \mid X_t) = P(R_t > u \mid X_t)$$

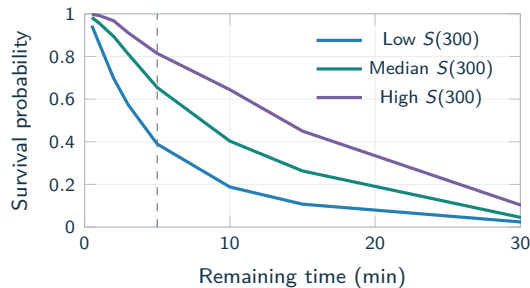
XGBoost-AFT uses survival bounds; the TCN predicts discrete hazards. Both retain censored samples.

Decision before shared post-processing

$$a_t = 1 \text{ if } S(300s \mid X_t) \geq 0.65 \\ \text{and } S_t \geq 10$$

0.65 is an exploratory test-selected threshold, not validation-calibrated.

Native test: TCN has lower IBS (0.103 vs 0.117); AFT higher switch F1 (0.863 vs 0.784).



Three actual TCN predictions at the 5th, 50th and 95th percentiles of predicted $S(300)$; display limited to 30 min.

Cross-Task Switch Performance

Selected representatives on the common reference grid

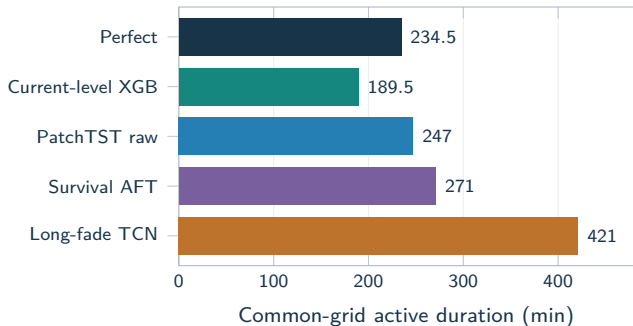
Representative	Coverage	Precision ↑	Recall ↑	F1 ↑
Current-level XGBoost	89.0%	1.000	0.808	0.894
Survival XGBoost-AFT	9.2%	0.812	0.938	0.870
PatchTST raw	89.2%	0.779	0.821	0.800
Long-fade TCN	28.7%	0.557	1.000	0.715

Missing native decisions map to zero. Coverage measures the fraction of reference timestamps with a native decision.

Survival threshold: exploratory 0.65. Descriptive test comparison, not a validation-selected cross-task winner.

Conservative activation, greater recovery and broad recall are different trade-offs.

Active Duration and Switching Events



Duration is not fragmentation

The same active time can be split into very different numbers of switch intervals.

Within-task example

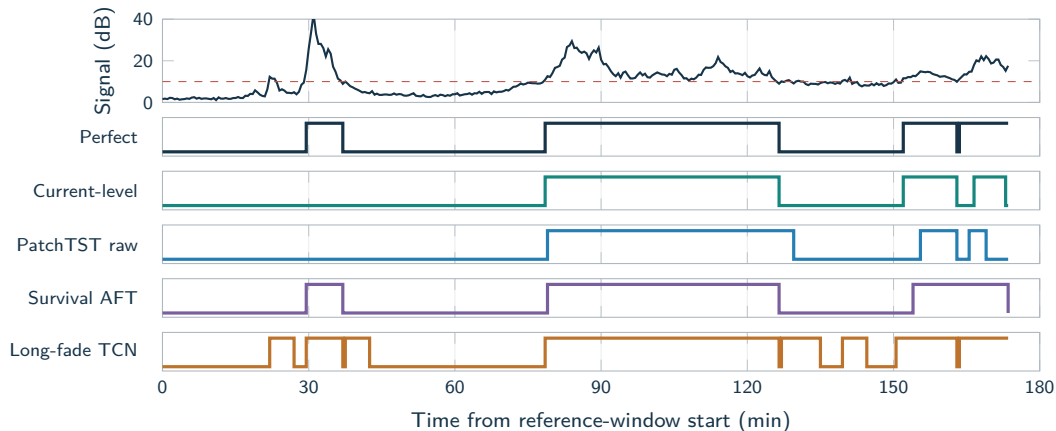
Native autoregressive grid:

Perfect: **18 intervals**

PatchTST: **20 intervals**

An interval is a contiguous run of switch 1, not necessarily a distinct physical fade. Native counts are not cross-task counts.

An Event-Level Comparison



Event dataset_007_event_00007. Final decisions: low = 0, high = 1. Missing native decisions map to 0; shared Perfect reference.

Limitations and Next Steps

Experimental limitation

Next step

One external test source

More terminals and independent acquisition periods

Exploratory survival threshold

Validation-only, cost-aware probability calibration

Offline evaluation and partial coverage

Streaming prototype with an explicit fallback policy

A deployed controller also needs a data-availability audit, persistent switch state, latency measurements and comparison against the engineering baseline.

These are offline experimental results, not evidence of deployment readiness.

Conclusions

1 Task definition changes the switch behavior.

A future trajectory, a level duration and an event label answer different questions.

2 Native predictive scores do not determine operational quality.

Evaluate timing, false activation, missed activity and decision coverage.

3 Tabular baselines remain competitive.

Current-level XGBoost and survival XGBoost-AFT expose useful operating trade-offs.

Evaluate the prediction and the decision policy together.

Thank you. Questions?